

Ex.no:6c

DATA VISUALIZATION USING POWER BI (HOTEL BOOKING)

Aim:

To clean, preprocess, and transform the raw **Life Expectancy dataset**, load the clean data model into Power BI, and build an interactive visual dashboard to analyze **life expectancy trends, country status, mortality, GDP, population, and schooling**.

Algorithm:

Step 1: Import Dataset: Load Life Expectancy Data.csv into the Power BI environment.

2. Deduplication:

Identify and remove duplicate rows across all fields.

3. Remove Unwanted Spaces:

Remove unnecessary spaces from column names such as BMI, HIV/AIDS, and thinness 1-19 years.

4. Handle Missing Values:

- Replace missing values in numerical columns such as **Life expectancy, Adult Mortality, GDP, Population, and Schooling** with suitable values such as median.
- Handle missing categorical values appropriately.

5. Handle Data Types:

- Format **Country** and **Status** as Text.
- Format **Year** as Whole Number.
- Format **Life expectancy, GDP, Schooling, and other numerical attributes** as Decimal Number or Whole Number as appropriate.

6. Load Clean Data:

Load the cleaned Life Expectancy table into Power BI Desktop using **Get Data** → **Text/CSV** and select **Transform Data** or **Load**.

7. Create Core Measure:

Right-click the Life Expectancy table and select **New Measure**.

Define the measure:

Average Life Expectancy =

AVERAGE('Life Expectancy Data'[Life expectancy])

8. Create Calculated Column:

Create a category based on life expectancy:

Life Expectancy Category =

IF(

```

'Life Expectancy Data'[Life expectancy] >= 70,
"High",
"Low"
)

```

9. Create Visualizations:

Create visualizations to show:

- Average Life Expectancy by Year
- Average Life Expectancy by Country Status
- Country-wise Life Expectancy
- GDP and Life Expectancy relationship
- Schooling and Life Expectancy relationship
- Distribution of Developed and Developing countries

10. Add Slicers:

Add slicers for **Country, Year, Status, and Life Expectancy Category** to enable interactive filtering.

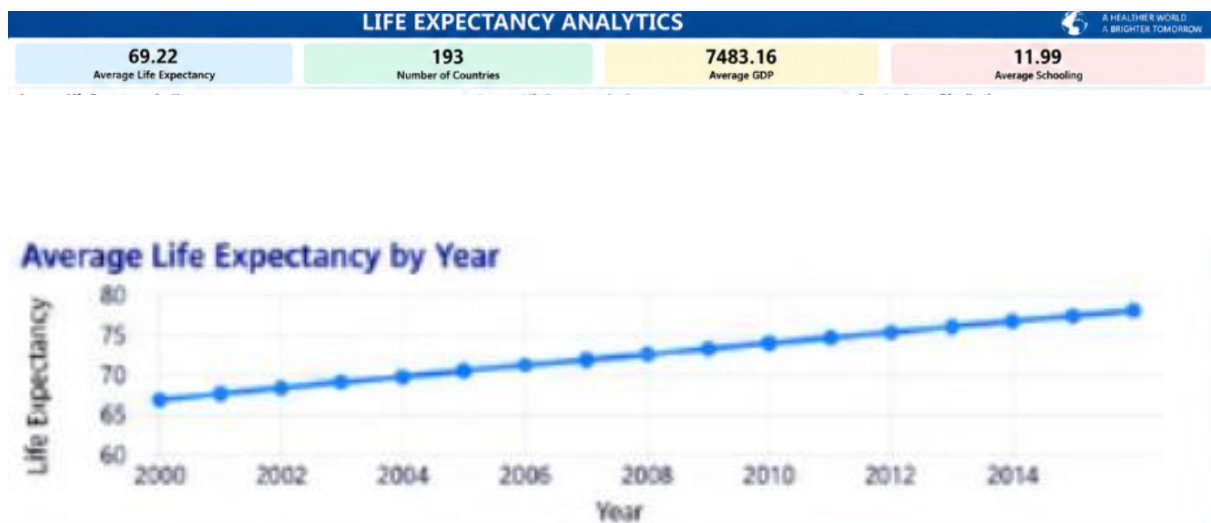
11. Create Dashboard:

Arrange cards, charts, graphs, and slicers into an interactive **Life Expectancy Analytics Dashboard**.

12. Save the Power BI file:

Save the completed dashboard as a **.pbix** file.

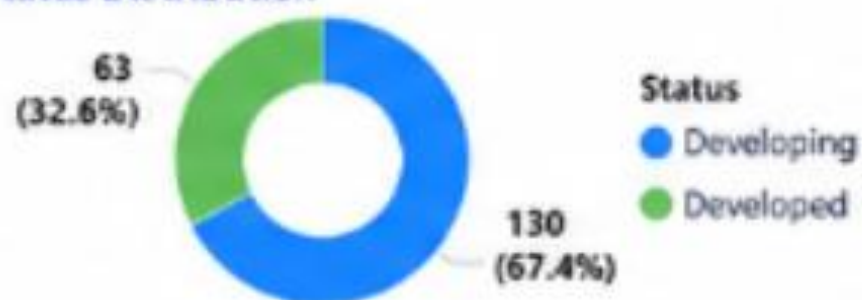
Output:



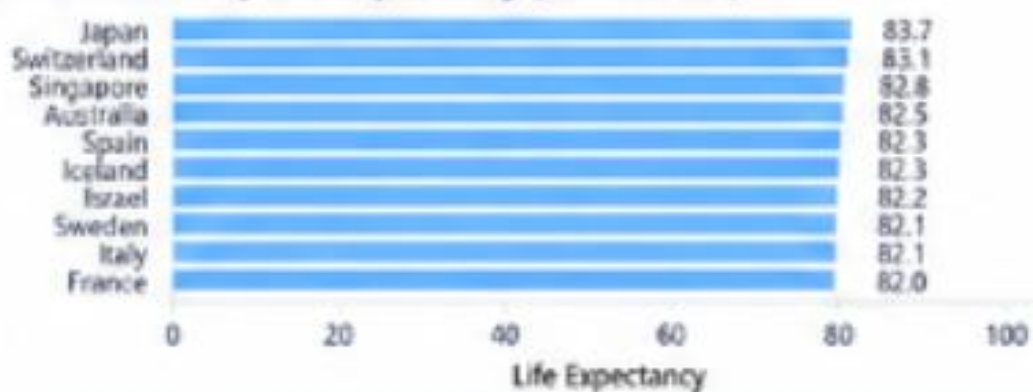
Average Life Expectancy by Status



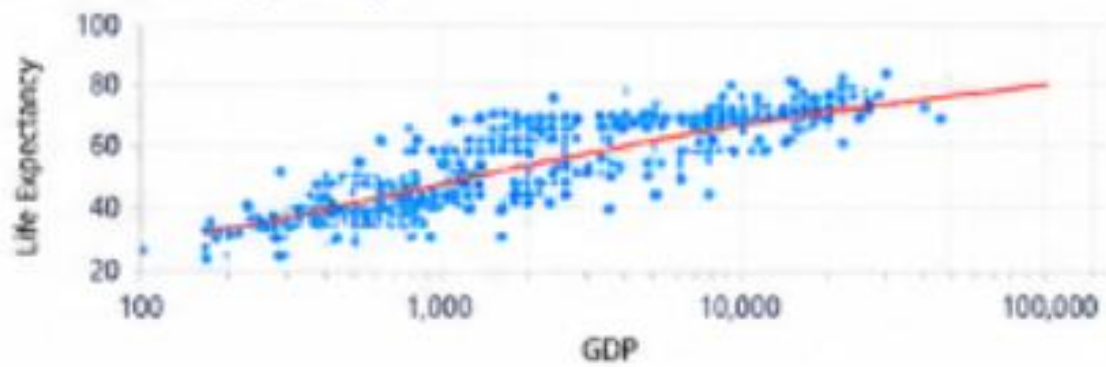
Country Status Distribution



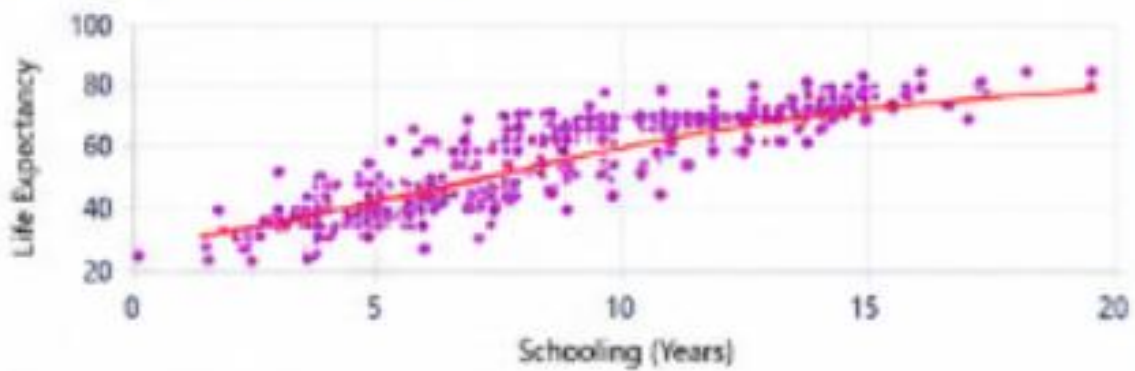
Top 10 Countries by Life Expectancy (Latest Year)



GDP vs Life Expectancy



Schooling vs Life Expectancy



Result: